is hard to find the best value for epsilon, the radius of the core-region determining what points are reachable from each other; when a dataset has areas with different densities, then it becomes even impossible to find a value that works equally well for the whole dataset (as shown in figure 12.14).

It doesn't matter if we try to search for this value manually or semi-automatically (see section 12.5), if we stick with DBSCAN algorithm alone, the algorithm is not enough to handle non-uniform datasets.

It didn't take too long until computer scientists came up with a new idea that could help in these situations: the key step missing was "**o**rdering **p**oints **t**o **i**dentify the **c**lustering **s**tructure", which the authors turned into an acronym, *OPTICS*[19], to name the algorithm they had invented.

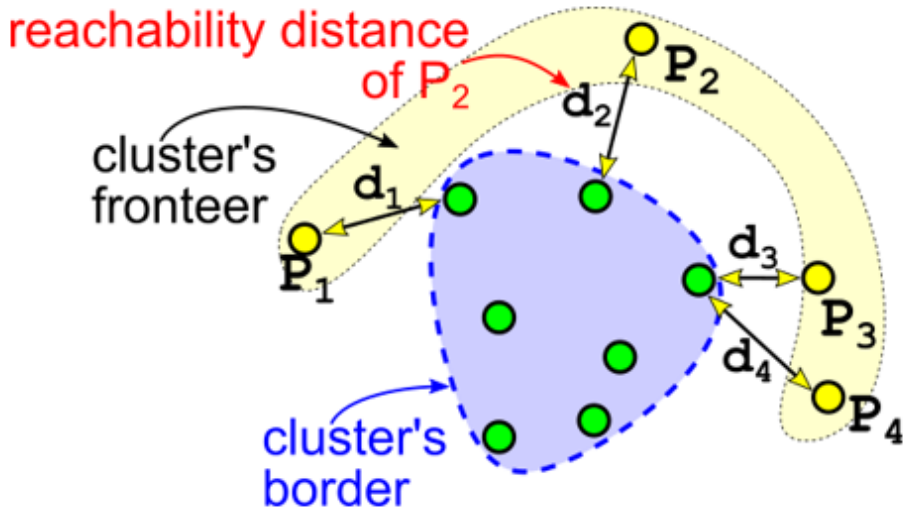


Figure 12.15 Considering the distance of undiscovered points (P_1 to P_4) from (points within) a certain cluster. The key idea in OPTICS is "discovering" these points from the closest to the